

Deep Speech Processing: Course Plan

S. R. Mahadeva Prasanna

Professor, Dept of DS & AI
Indian Institute of Information Technology Dharwad

prasanna@iiitdwd.ac.in



December 30, 2024

Some Resources for Deep Speech Processing

- L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004
- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
- R. Prabhavalkar, et.al., "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024,
- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." (2024).
- Wang, S., et.al., "Overview of speaker modeling and its applications: From the lens of deep speaker representation learning" IEEE/ACM Transactions on Audio, Speech, and Language Processing.

Theory Course Information

- Course Title : Deep Speech Processing (DSP)
- Course Code : DS XXX
- Venue : C305 (Mode of delivery hybrid, students residing outside campus / out station can attend online, however, keep camera on during attending classes)
- Mon : 630-8.00 pm
- Tue : 630-8.00 pm
- Thu : 6.30-8.00 pm (extra slot, when needed only, please keep this slot free for yourself)



Theory Course (3-0-0-4-4) - Team

- Course Instructor & Coordinator: S R M Prasanna
- Co-Instructors:
 - Dr. Nagaraj Adiga (Associate Director, Krutrim, Bengaluru);
 - Dr. Sishir Kalita (Principal Data Scientist, Armsofttech, Chennai);
 - Dr. Mrinmoy Bhattacharjee (Asst. Professor, Dept of CSE, IIT Jammu);
 - Dr. Jagabandhu Mishra (Postdoc, Univ. of Eastern Finland))
 - Course TAs: Rishith Sadashiv & Jyoti Gadad



Course Evaluation (3-0-0-4-4)

- First instructional day (T_0)
- Except endsem, outstation candidates can take all other components of evaluation in online mode
- Quiz-I (5%) : $T_0 + 4W$ Thursday, 23rd Jan 25, 6.30-7.30 pm
- Midsem (15%) : As per calendar, $T_0 + 7W$;
- Quiz-II (5%): $T_0 + 11W$, Thursday, 20th Mar 25, 6.30-7.30 pm
- Endsem (25%): As per calendar (Offline for all), $T_0 + 14W$;
- Lab implementation (15%) & Course Project (15%) (CE)
- Surprise evaluation (SE) (10%) & Endsem viva-voce (10%)
- 50% by written exams + 50% by other components
- Attendance Policy : As per institute norms

- **Overview of Speech Processing:** Speech production, perception and comprehension, spoken language processing, human speech processing tasks, nature of speech signal, signal processing of speech, linguistics and phonetics, feature extraction. (6 hrs)
- **Machine Learning of Speech Processing:** clustering of speech features, classification tasks in speech, Gaussian mixture model (GMM), Hidden Markov Model (HMM), Support Vector Machine (SVM) and Neural Networks (NN) (6 hrs)
- **Deep Learning of Speech Processing:** Self-Supervised Learning for Speech Representation. Automatic Speech Recognition (ASR). Text to Speech (TTS) synthesis. Speaker and Language Recognition. (Details in Annexure - DSP) (30 hrs)

Self-Supervised Learning (SSL) for Speech Representation

- Representation learning for speech: a brief historical context.
- Self-Supervised Learning (SSL): Motivation: reducing labeled data requirements, leveraging large unlabeled corpora. General frameworks: contrastive, generative, and predictive.
- Pretraining: Mockingjay, wav2vec 2.0, HuBERT, etc.
- Fine-tuning and domain adaptation: Fine-tuning SSL models for downstream tasks, resource-constrained adaptation (low-resource languages, few-shot learning)
- Benchmarking: To understand information encoded in speech representations and to, evaluate and compare their performance. For example, SUPERB, IndicSUPERB, etc.



References for SSL

- Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
- Abdelrahman Mohamed, et al., Self-Supervised Speech Representation Learning: A Review, arxiv.2205.10643
- A. Baevski, et al., “wav2vec 2.0: A framework for self-supervised learning of speech representations,” Proc. of NeurIPS, vol. 33, 2020.
- W.-N. Hsu, et al., “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” arXiv preprint arXiv:2106.07447, 2021.
- Javed, Tahir, et al. “Indicsuperb: A speech processing universal performance benchmark for Indian languages.” Proceedings of AAAI Conference on Artificial Intelligence. Vol. 37. No. 11. 2023.
- Arora, Siddhant, et al. “On the Evaluation of Speech Foundation Models for Spoken Language Understanding.” arXiv preprint

Automatic Speech Recognition (ASR)

- Introduction and brief history, types of ASR systems (Cascaded, End-to-End), evaluation metric - WER, CER
- Cascaded ASR system - acoustic model (GMM-HMM, DNN-HMM, TDNN), language models (n-gram, RNN), WFST decoding, greedy decoding, beam search decoding
- E2E ASR systems - Encoder-Decoder, CTC, RNN-T; ASR using LLMs
- Streaming vs. Non-Streaming ASR, Popular ASR systems - Whisper, wav2vec2.0; Variants of Transformers; Popular datasets, Toolkits, Open problems
- Introduction to Mamba for ASR



References for ASR

- Gales, M., Young, S. (2008). The application of hidden Markov models in speech recognition. Foundations and Trends® in Signal Processing, 1(3), 195-304.
- Alex Graves, Navdeep Jaitly, "Towards End-to-End Speech Recognition with Recurrent Neural Networks", Proc 31st Int. Conf. on Machine Learning, PMLR 32(2):1764-1772, 2014. <http://proceedings.mlr.press/v32/graves14.pdf>
- CS753, IITB, Instructor: Dr. Preethi Jyothi. Course website: <https://www.cse.iitb.ac.in/~pjyothi/cs753/index.html>
- Malik, M., Malik, M. K., Mehmood, K., Makhdoom, I. (2021). Automatic speech recognition: a survey. Multimedia Tools and Applications, 80, 9411-9457.
- R. Prabhavalkar, et. al., "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024, doi: 10.1109/TASLP.2023.3328283.
- Speech Recognition — Deep Speech, CTC, Listen, Attend, and Spell:
<https://jonathan-hui.medium.com/speech-recognition-deep-speech-ctc-listen-attend-and-spell-d05e940e9ed1>



Text to Speech Synthesis (TTS)

- Introduction: Traditional methods, Cascaded methods, End-to-End Methods, Applications, Evaluation Criteria
- Components of TTS: Text front end, Acoustic model and Vocoder
- Advanced TTS Systems: Transformer, GAN, Flow, and Diffusion based models
- Developing TTS systems: Data collection procedure, Cleaning text/audio, Demo/training using toolkit and Hands-on exploration
- Future perspective: Voice cloning under limited data, Conversational/emotional TTS, adherence to ethical rules



References for TTS

- Triantafyllopoulos, Andreas, et al. "An overview of affective speech synthesis and conversion in the deep learning era." Proceedings of the IEEE 111.10 (2023): 1355-1381.
- Zhang, Chenshuang, et al. "A survey on audio diffusion models: Text to speech synthesis and enhancement in generative ai." arXiv preprint arXiv:2303.13336 (2023)
- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." arXiv preprint arXiv:2412.06602 (2024).
- Tan, Xu, et al. "A survey on neural speech synthesis." arXiv preprint arXiv:2106.15561 (2021).
- Mu, Zhaoxi, Xinyu Yang, and Yizhuo Dong. "Review of end-to-end speech synthesis technology based on deep learning." arXiv preprint arXiv:2104.09995 (2021).

Speaker and Language Recognition (SLR)

- Speaker information in his speech: physiological and behavioral, well-known features: MFCC, Prosody, excitation, speech representations: i-vector, d-vector, x-vector (TDNN, ECAPA-TDNN, ResNet34), wavlm etc., generative and discriminative strategy, backend classifiers: Cosine, PLDA, Neural PLDA, Neural SVM, performance metrics: EER and detection cost function (DCF), Calibration and Fusion techniques.



References for SLR

- Wang, S., Chen, Z., Lee, K. A., Qian, Y., Li, H. (2024). Overview of speaker modeling and its applications: From the lens of deep speaker representation learning. IEEE/ACM Transactions on Audio, Speech, and Language Processing.
- Sharma, R., Govind, D., Mishra, J., Dubey, A. K., Deepak, K. T., Prasanna, S. R. M. (2024). Milestones in speaker recognition. Artificial Intelligence Review, 57(3), 58.
- Kinnunen, T., Li, H. (2010). An overview of text-independent speaker recognition: From features to supervectors. Speech communication, 52(1), 12-40.
- Li, H., Ma, B., Lee, K. A. (2013). Spoken language recognition: from fundamentals to practice. Proceedings of the IEEE, 101(5), 1136-1159.

Lab Implementation

- Individual
- Some 10 implementations of different speech processing concepts
- Demonstrating the understanding of working principle
- Nature of digital speech
- Speech signal processing
- Machine learning of speech
- Deep learning of speech



Course Project

- Team of 1-2 members
- Fair policy of declaration of contribution
- Speech processing research based projects
- Problem titles will be floated
- Discuss and choose one
- Working with my past and present research scholars, and some collaborators



Quiz, Midsem, Endsem

- Written exam (1, 2, 3 hr)
- Descriptive, derivation, problem solving
- Not direct questions
- Mind twisters



History of This Course

- **Ver 1.0:** Speech Technology
- **Ver 2.0:**
 - Speech Signal Processing & Coding
 - Speech Technology
- **Ver 3.0:**
 - Speech Processing
 - Advanced Topics in Speech Processing
- **Ver 4.0:** Deep Speech Processing

